INTEGRATED INVERTER / STM32G431 / REV B REVIEW

One PCB / one MCU / one three-phase bridge / one external-resistor brake.
12-60 V bus, DC input <=10 A; phase 10 Arms design target; bridge PWM 20 kHz; separate 12-24 V auxiliary.

Auxiliary input

eFuse / 12-24 V

文件: aux_input.kicad_sch

Driver supply

12 V buck-boost

文件: supply_12v.kicad_sch

Logic and sensors

5 V / 3.3 V / supervisor

文件: supply_logic.kicad_sch

Bus and brake

12-60 V / external resistor

文件: bus_brake.kicad_sch

U half bridge

TO-263 / driver / shunt

文件: bridge_u.kicad_sch

V half bridge

TO-263 / driver / shunt

文件: bridge_v.kicad_sch

W half bridge

TO-263 / driver / shunt

文件: bridge_w.kicad_sch

Current sensing

3 x INA241 / 1.65 V bias

文件: current_sense.kicad_sch

Voltage and NTC

Bus / 3 phases / NTC

文件: voltage_sense.kicad_sch

Analog protection

Current / bus OV / driver UV

文件: comparators.kicad_sch

Hardware latch

Healthy chain / reset / fault

文件: fault_logic.kicad_sch

Hardware PWM gates

Six independent AND gates

文件: control_interface.kicad_sch

Local STM32G431

LQFP64 / VDD / pin plan

文件: mcu.kicad_sch

MCU analog inputs

8 RC filters / VDDA / VREF

文件: mcu_analog.kicad_sch

Encoders and SPI

2 x ABZ / 2 x SPI

文件: sensors.kicad_sch

Debug and communication

SWD / TTL / optional CAN / LEDs

文件: debug_comms.kicad_sch

Local command coupling

PWM / enable / fault / HSE option

文件: mcu_control.kicad_sch

Native Rev B drawing is for circuit review; PCB placement/routing and rated operation have not been validated.
Independent comparators, latch and PWM gates retained; MCU added locally. No 24-pin controller cable or external brake-command cable.
No main-bus precharge/bleeder, USB, Hall or PFC/LLC power circuit. PFC power-board drawing is paused.
3.3V single-ended encoder/SPI interfaces; exact external modules, connectors, passives, thermal and protection dynamics still need review.

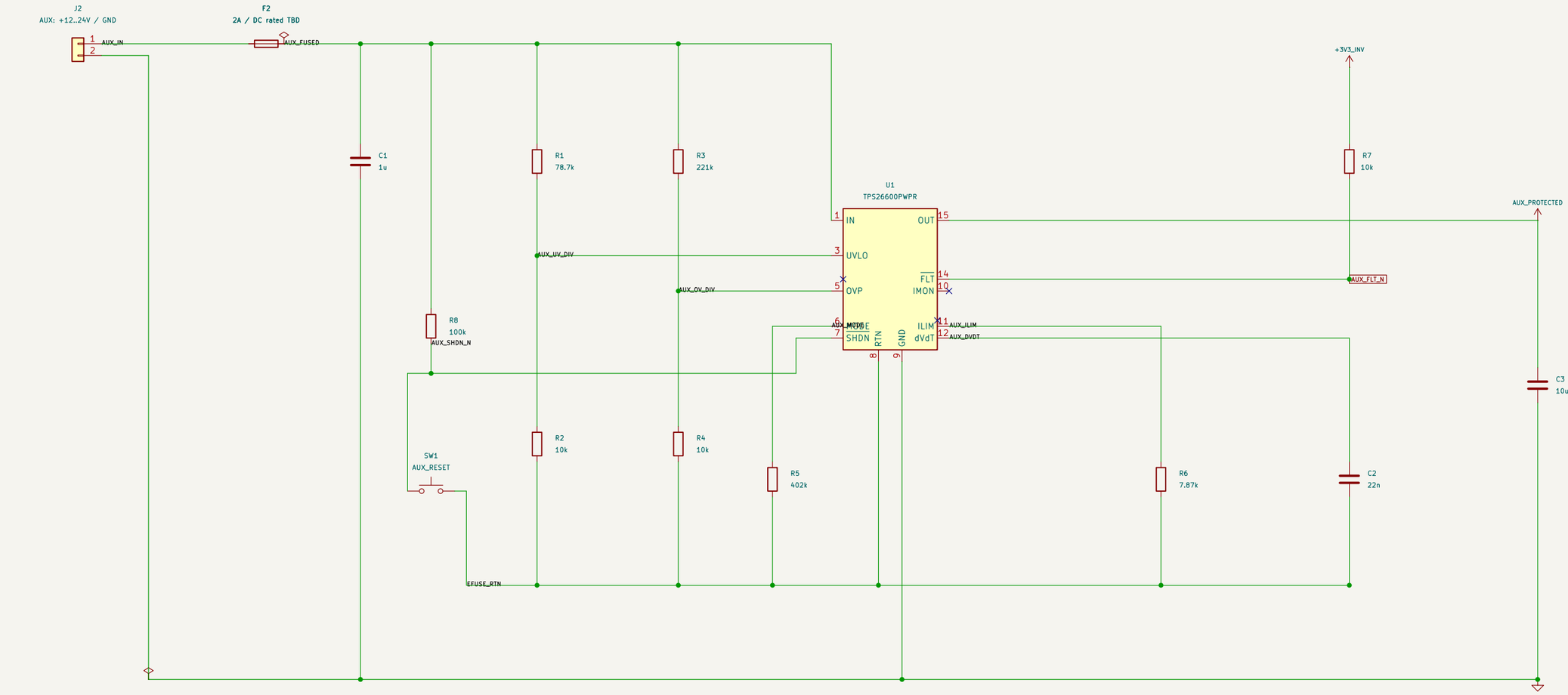
INVERTER | non-isolated laboratory draft | 12-60 V bus | separate 12-24 V auxiliary
HARDWARE REVIEW DRAFT — NOT RELEASED FOR PCB
power_system

Sheet: /
File: inverter.kicad_sch

Title: Inverter — integrated STM32G431 review draft

Size: A3	Date: 2026-11-06	**Rev: B-integrated-draft**
KiCad E.D.A. 10.0.7		Id: 1/18

AUXILIARY INPUT / eFUSE / 12–24 V



RTN / exposed pad is the eFuse reference island. It is NOT the board GND pin.
MODE=402k selects latch-off. AUX_RESET resets the auxiliary path; no MCU startup dependency.
UV / OV / current-limit values remain unchanged; dynamics and surge coordination still require review.

INVERTER | non-isolated laboratory draft | 12–60 V bus | separate 12–24 V auxiliary
HARDWARE REVIEW DRAFT – NOT RELEASED FOR PCB

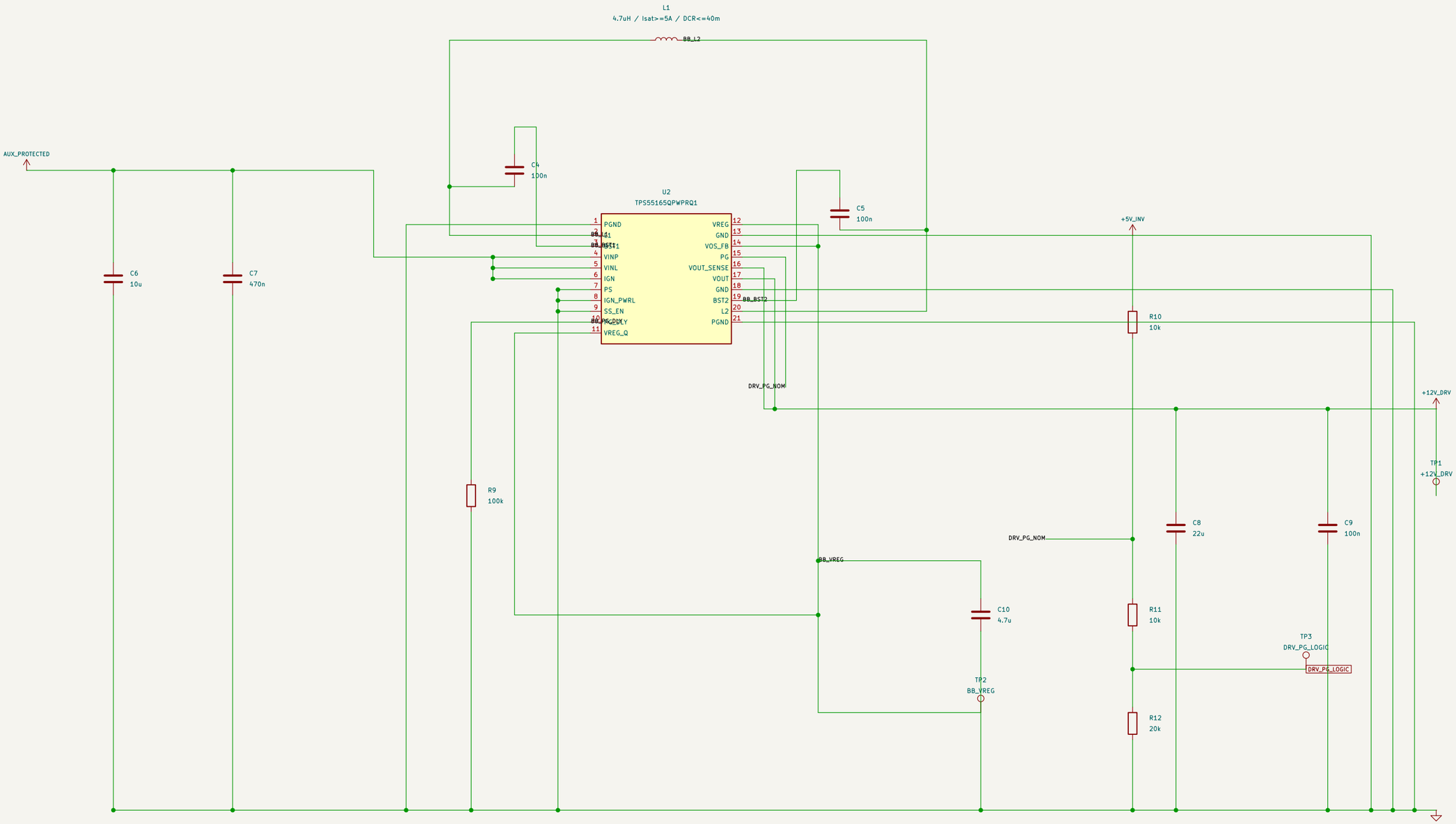
power_system

Sheet: /Auxiliary input/
File: aux_input.kicad_sch

Title: Inverter – 12–24 V auxiliary input protection

Size: A3	Date: 2026-11-06	**Rev: B-readable-draft**
KiCad E.D.A. 10.0.7		Id: 2/18

12 V DRIVER RAIL / TPS55165 BUCK-BOOST



VREG_Q and VOS_FB return to VREG; VREG is not a load rail. VOUT_SENSE is tied to the 12 V output.
Input/output/bootstrap wiring is shown directly. PG high is about 2.5 V with the fitted pullup/divider.
Inductor model, startup, worst-case gate voltage and thermal capability remain review items.

INVERTER | non-isolated laboratory draft | 12-60 V bus | separate 12-24 V auxiliary
HARDWARE REVIEW DRAFT - NOT RELEASED FOR PCB

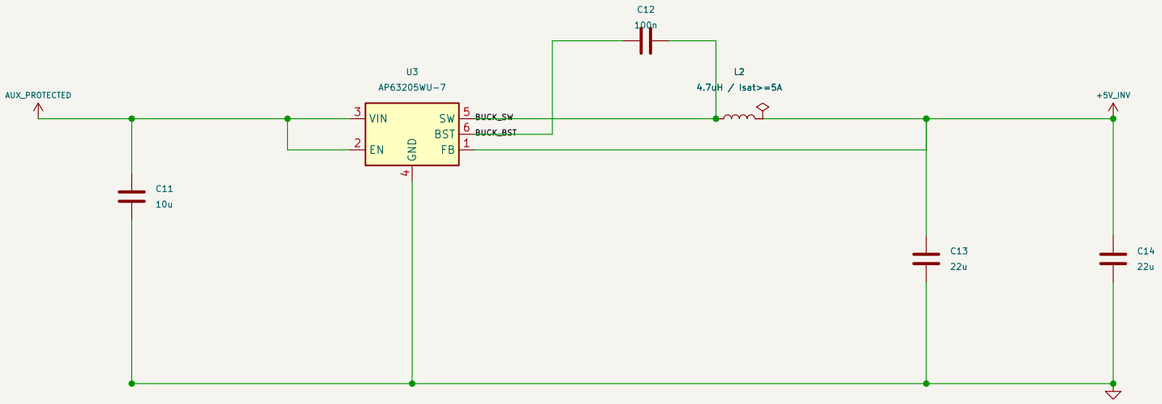
power_system

Sheet: /Driver supply/
File: supply_12v.kicad_sch

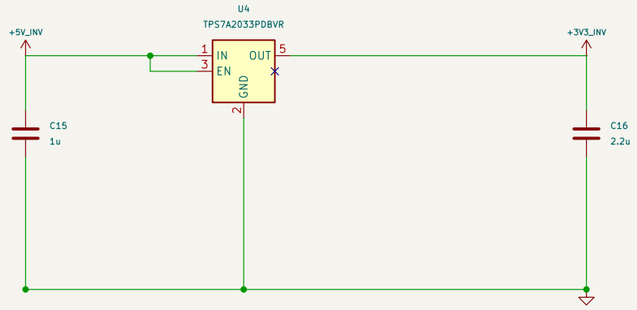
Title: Inverter - regulated 12 V driver supply

Size: A3	Date: 2026-11-06	**Rev: B-readable-draft**
KiCad E.D.A. 10.0.7		Id: 3/18

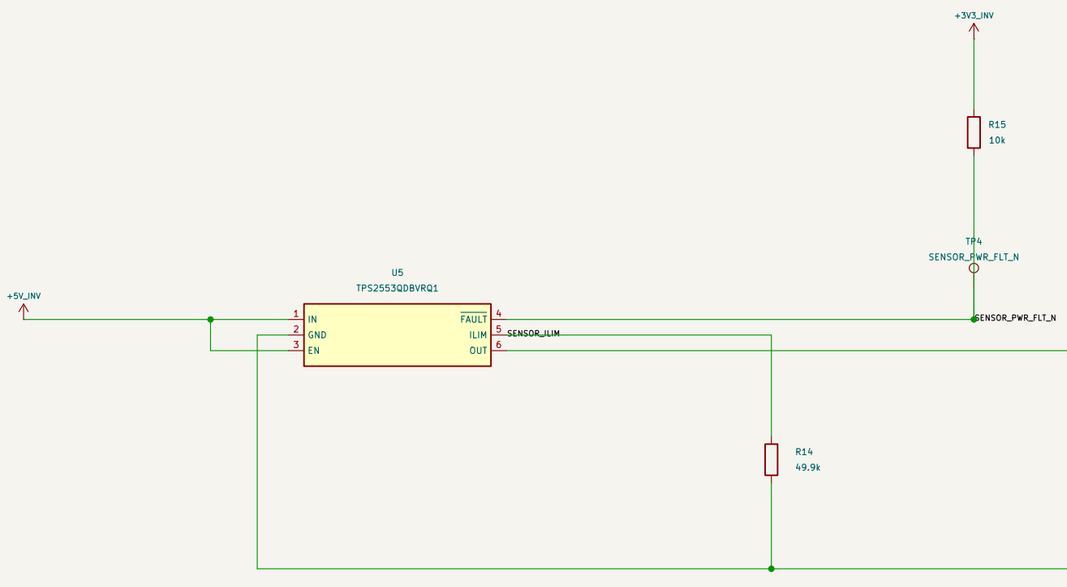
5 V BUCK / AP63205



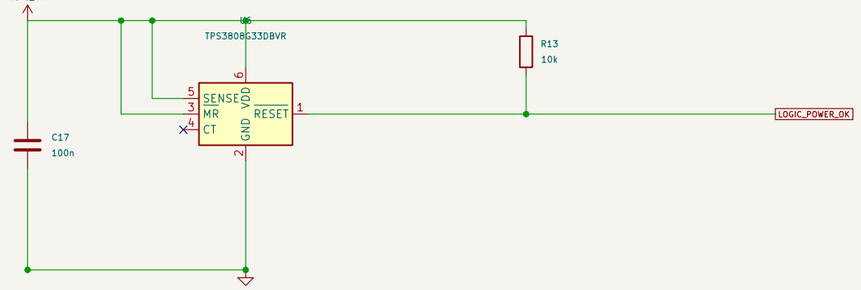
3.3 V MCU / ANALOG SUPPLY



PROTECTED OPTIONAL SENSOR POWER / TOTAL TARGET 0.5 A



3.3 V SUPERVISOR / LOCAL MCU RESET



J7 / external power links default DNP. The branch current target is total, not per connector.
The 3.3 V LDO is shared by MCU, analog, logic and optional CAN; budget and temperature still require verification.

INVERTER | non-isolated laboratory draft | 12-60 V bus | separate 12-24 V auxiliary
HARDWARE REVIEW DRAFT - NOT RELEASED FOR PCB

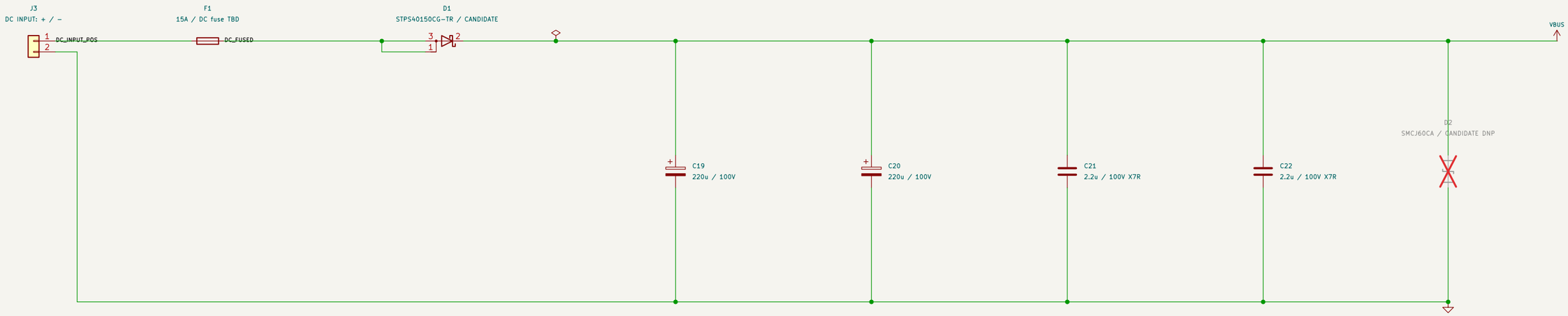
power_system

Sheet: /Logic and sensors/
File: supply_logic.kicad_sch

Title: Inverter - 5 V, 3.3 V and sensor power

Size: A3	Date: 2026-11-06	**Rev: B-readable-draft**
KiCad E.D.A. 10.0.7		Id: 4/18

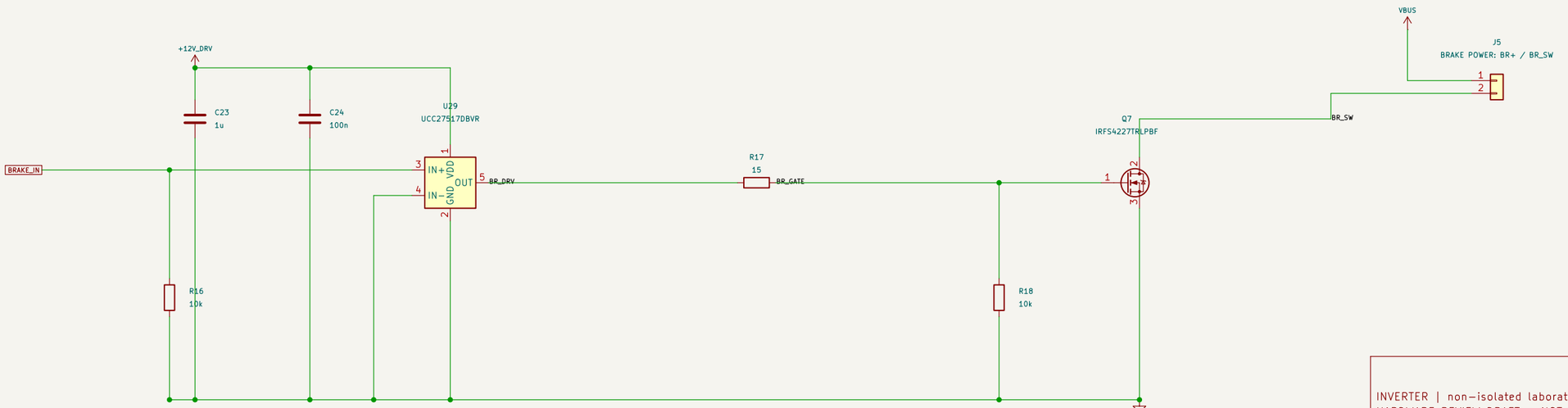
MAIN BUS INPUT / BULK CAPACITORS / RAW PWM OUTPUTS



D2 is DNP. The input diode / TVS / fuse are still procurement and energy/thermal review items.
No main-bus precharge or dedicated bleeder. U/V/W are raw bridge PWM, not LC-filtered sine outputs.



ONE BRAKE BRANCH / EXTERNAL RESISTOR / LOCAL MCU COMMAND



J5: BR+ = VBUS, BR_SW = MOS drain. External resistor goes across these pins; unused port stays OPEN.
Brake is independent of the bridge ENABLE/fault gates. MCU reset defaults the command low.

INVERTER | non-isolated laboratory draft | 12–60 V bus | separate 12–24 V auxiliary
HARDWARE REVIEW DRAFT – NOT RELEASED FOR PCB

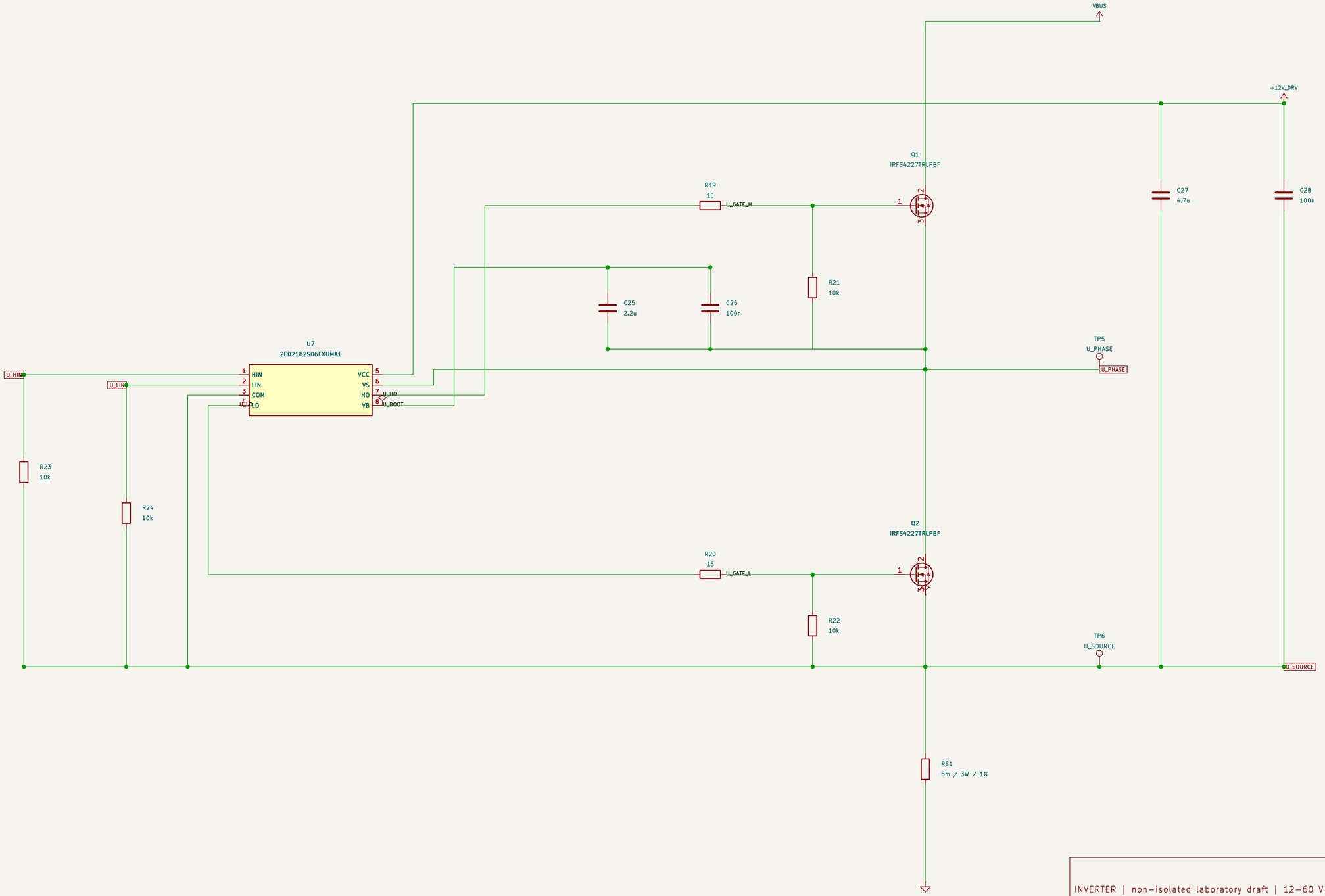
power_system

Sheet: /Bus and brake/
File: bus_brake.kicad_sch

Title: Inverter – DC bus input, bulk capacitors and brake

Size: A3	Date: 2026-11-06	**Rev: B-readable-draft**
KiCad E.D.A. 10.0.7		Id: 5/18

U HALF BRIDGE / VISIBLE GATE, BOOTSTRAP AND SHUNT RETURNS



COM, low-gate pulldown, local VCC capacitors and sense high all return to SOURCE above the shunt.
High-gate pulldown and bootstrap capacitors return to PHASE. Common GND/DC- exists only below RS.
Gate R / bootstrap values remain initial values; deadtime, minimum recharge time and waveforms require validation.

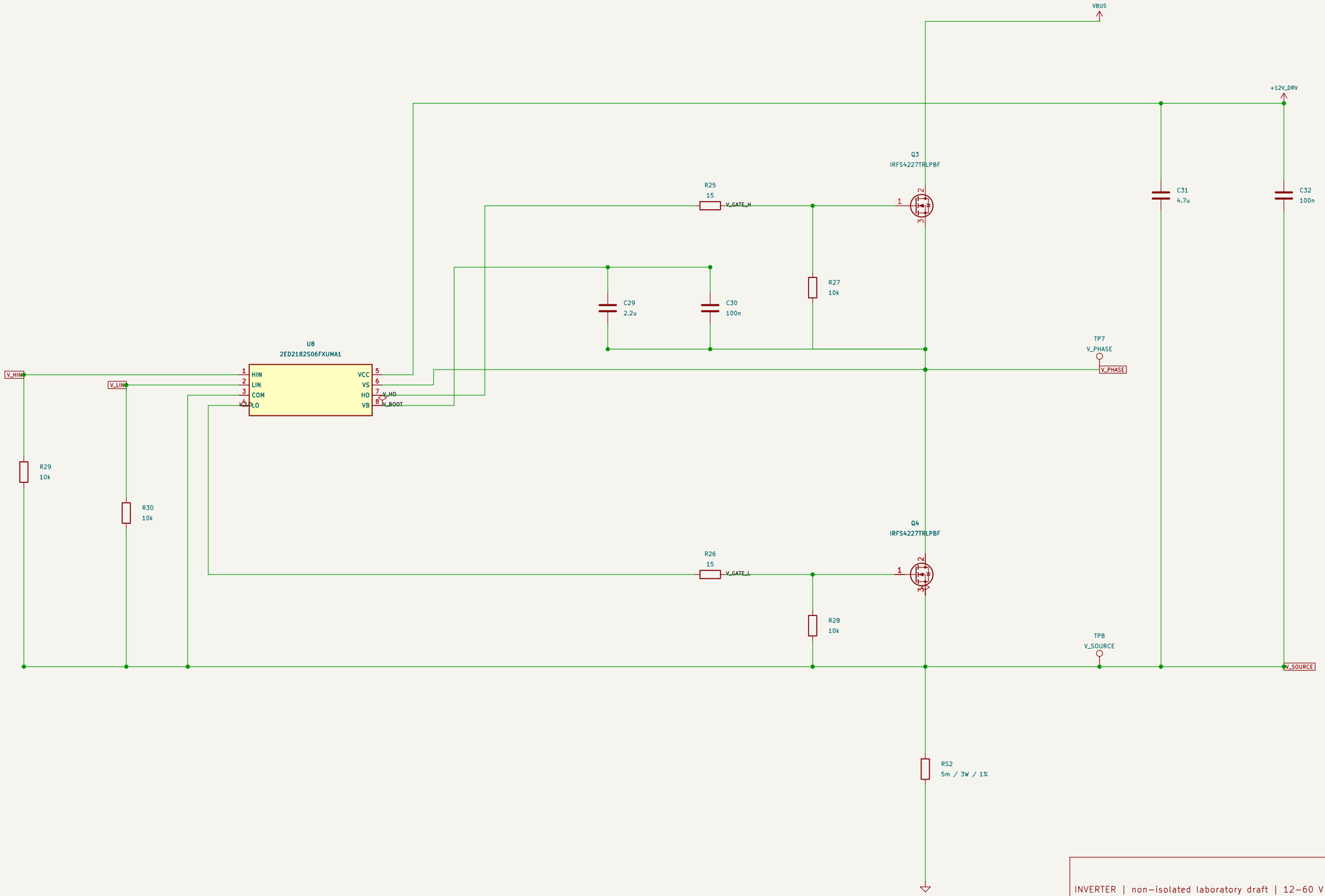
INVERTER | non-isolated laboratory draft | 12-60 V bus | separate 12-24 V auxiliary
HARDWARE REVIEW DRAFT - NOT RELEASED FOR PCB
power_system

Sheet: /U half bridge/
File: bridge_u.kicad_sch

Title: Inverter - U half bridge and low-side shunt

Size: A3	Date: 2026-11-06	**Rev: B-readable-draft**
KiCad E.D.A. 10.0.7		Id: 6/18

V HALF BRIDGE / VISIBLE GATE, BOOTSTRAP AND SHUNT RETURNS



COM, low-gate pulldown, local VCC capacitors and sense high all return to SOURCE above the shunt.
High-gate pulldown and bootstrap capacitors return to PHASE. Common GND/DC- exists only below RS.
Gate R / bootstrap values remain initial values; deadtime, minimum recharge time and waveforms require validation.

INVERTER | non-isolated laboratory draft | 12-60 V bus | separate 12-24 V auxiliary
HARDWARE REVIEW DRAFT - NOT RELEASED FOR PCB

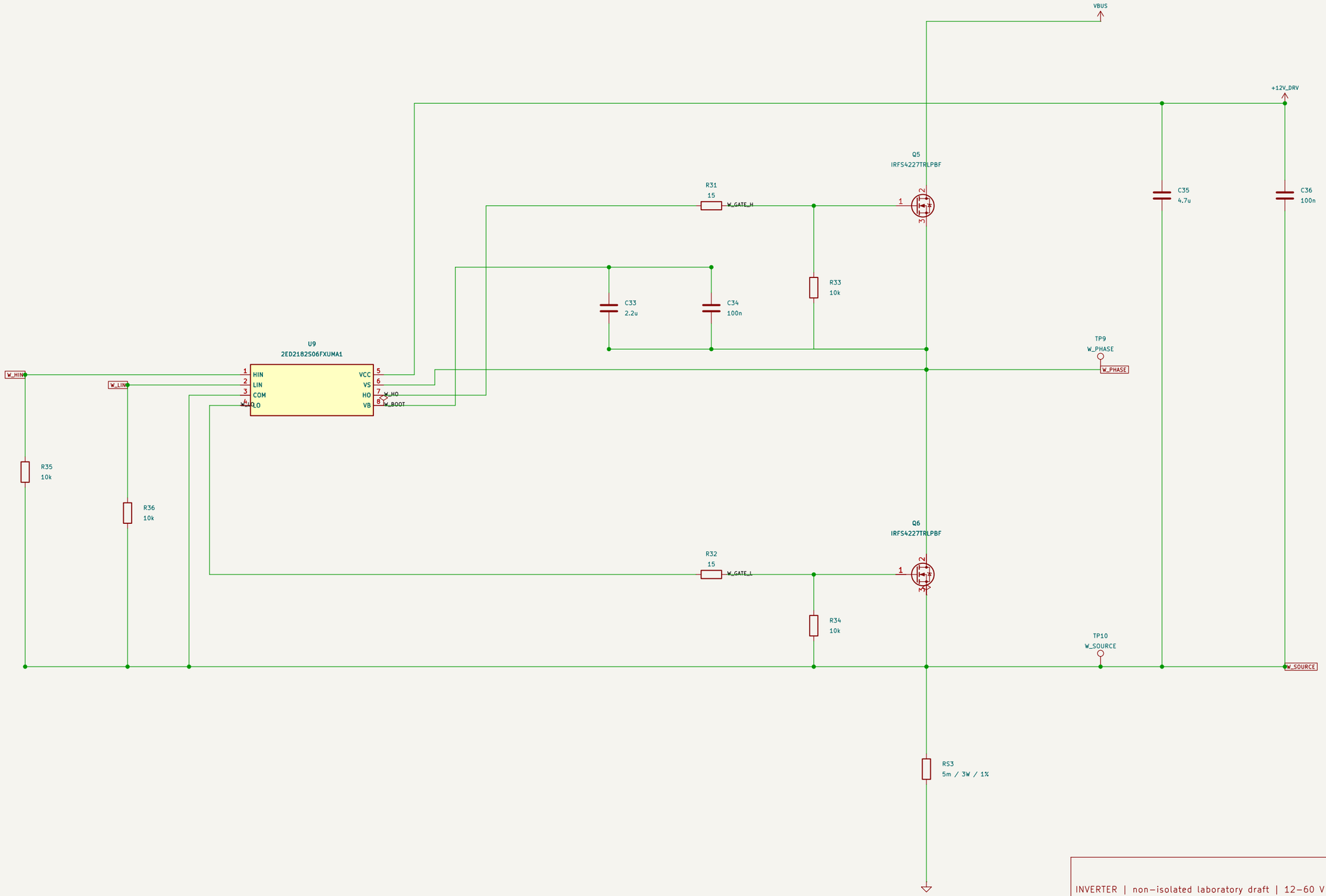
power_system

Sheet: /V half bridge/
File: bridge_v.kicad_sch

Title: Inverter - V half bridge and low-side shunt

Size: A3	Date: 2026-11-06	**Rev: B-readable-draft**
KiCad E.D.A. 10.0.7		Id: 7/18

W HALF BRIDGE / VISIBLE GATE, BOOTSTRAP AND SHUNT RETURNS



COM, low-gate pulldown, local VCC capacitors and sense high all return to SOURCE above the shunt.
High-gate pulldown and bootstrap capacitors return to PHASE. Common GND/DC- exists only below RS.
Gate R / bootstrap values remain initial values; deadtime, minimum recharge time and waveforms require validation.

INVERTER | non-isolated laboratory draft | 12-60 V bus | separate 12-24 V auxiliary
HARDWARE REVIEW DRAFT - NOT RELEASED FOR PCB

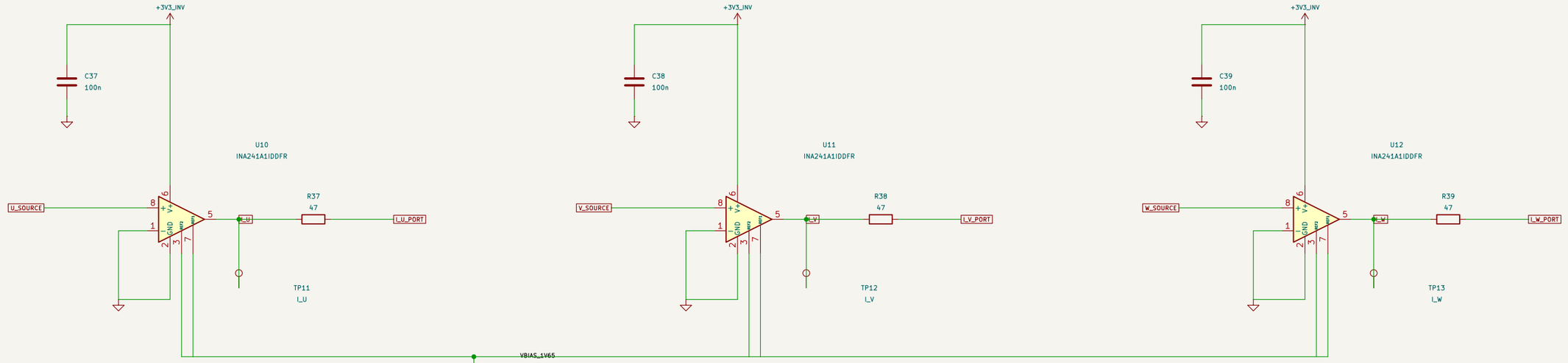
power_system

Sheet: /W half bridge/
File: bridge_w.kicad_sch

Title: Inverter - W half bridge and low-side shunt

Size: A3	Date: 2026-11-06	**Rev: B-readable-draft**
KiCad E.D.A. 10.0.7		Id: 8/18

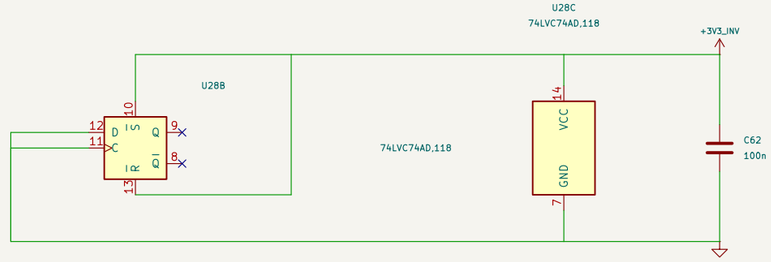
THREE LOW-SIDE KELVIN CURRENT CHANNELS



REF1 + REF2 share the buffered 1.65 V bias below.

BUFFERED BIAS AND CURRENT WINDOW REFERENCES

SPARE U28B / FIXED INPUTS



I = 1.65 V + 50 mV/A; nominal current trips +/−20 A. Low-side shunts have switching-state blind intervals.
References track the local 3.3 V rail; phase-current sign and sampling windows require bench validation.

INVERTER | non-isolated laboratory draft | 12–60 V bus | separate 12–24 V auxiliary
HARDWARE REVIEW DRAFT — NOT RELEASED FOR PCB

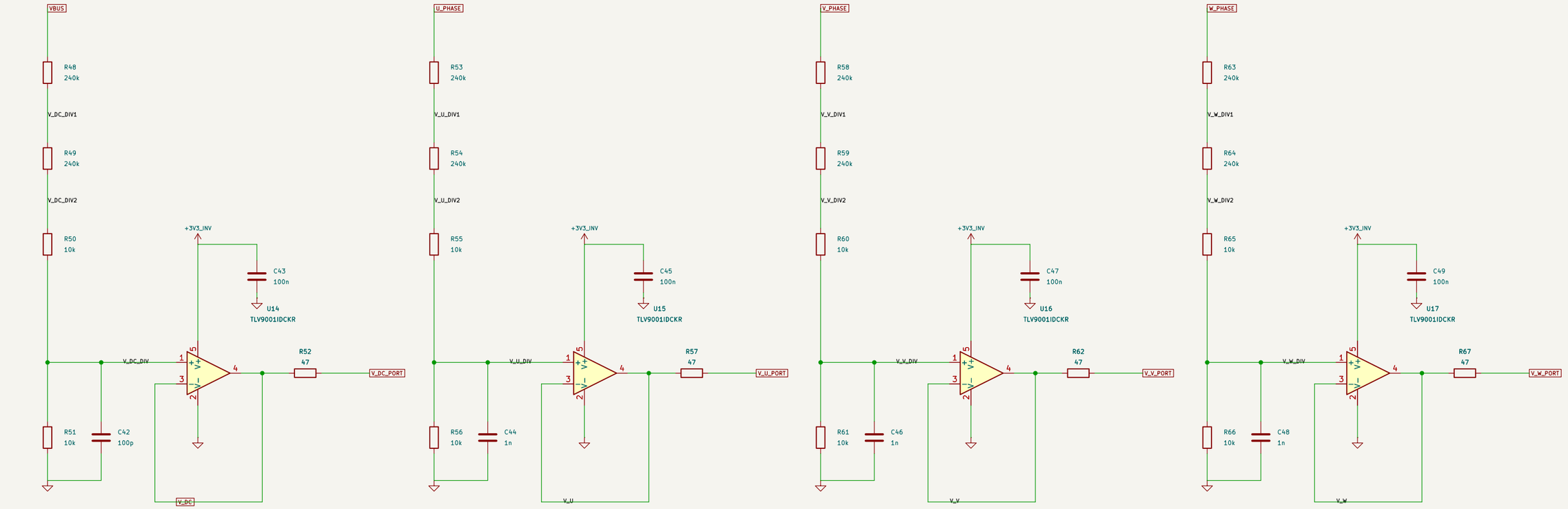
power_system

Sheet: /Current sensing/
File: current_sense.kicad_sch

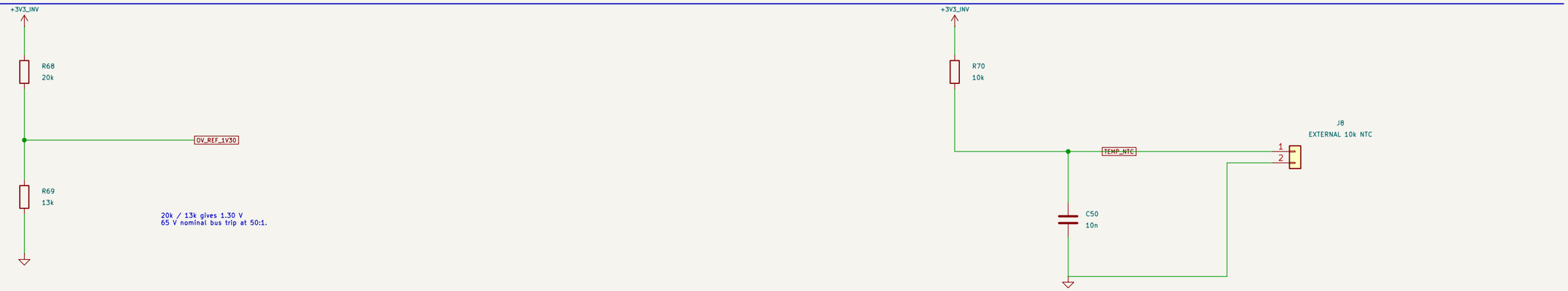
Title: Inverter — three current channels and references

Size: A3	Date: 2026-11-06	**Rev: B-readable-draft**
KiCad E.D.A. 10.0.7		Id: 9/18

FOUR 50:1 VOLTAGE CHANNELS / COMPLETE DIVIDER TO BUFFER



BUS OVERVOLTAGE REFERENCE / EXTERNAL NTC



Raw phase sense is measured before an external LC filter. Bus RC ~0.98 us; phase RC ~9.8 us.
External NTC nominal 10k; beta, physical placement and cable protection remain to be selected.

INVERTER | non-isolated laboratory draft | 12-60 V bus | separate 12-24 V auxiliary
HARDWARE REVIEW DRAFT - NOT RELEASED FOR PCB

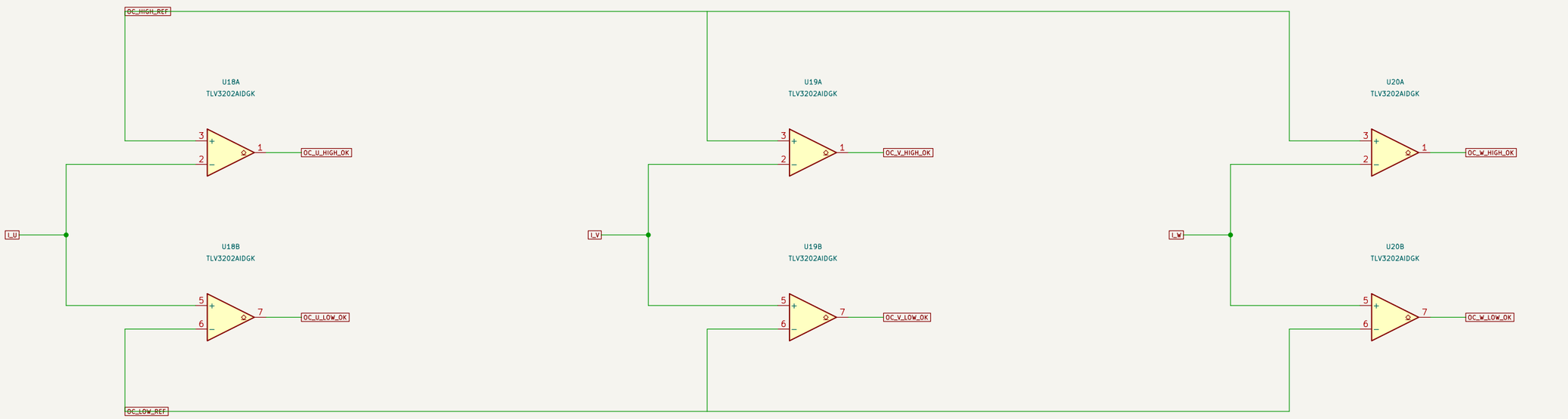
power_system

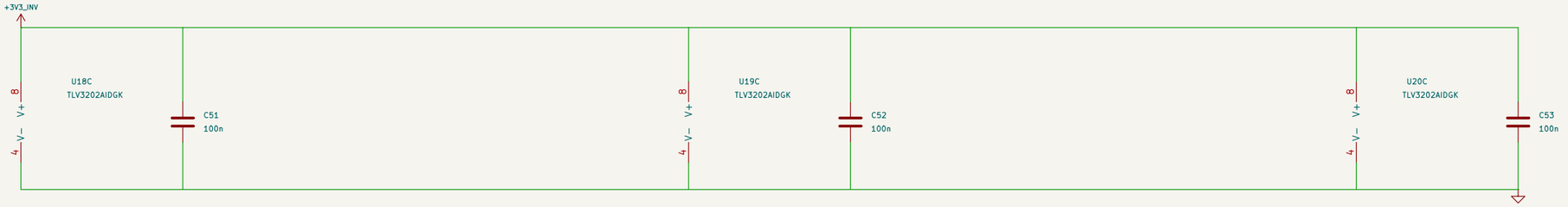
Sheet: /Voltage and NTC/
File: voltage_sense.kicad_sch

Title: Inverter - voltage and NTC sensing

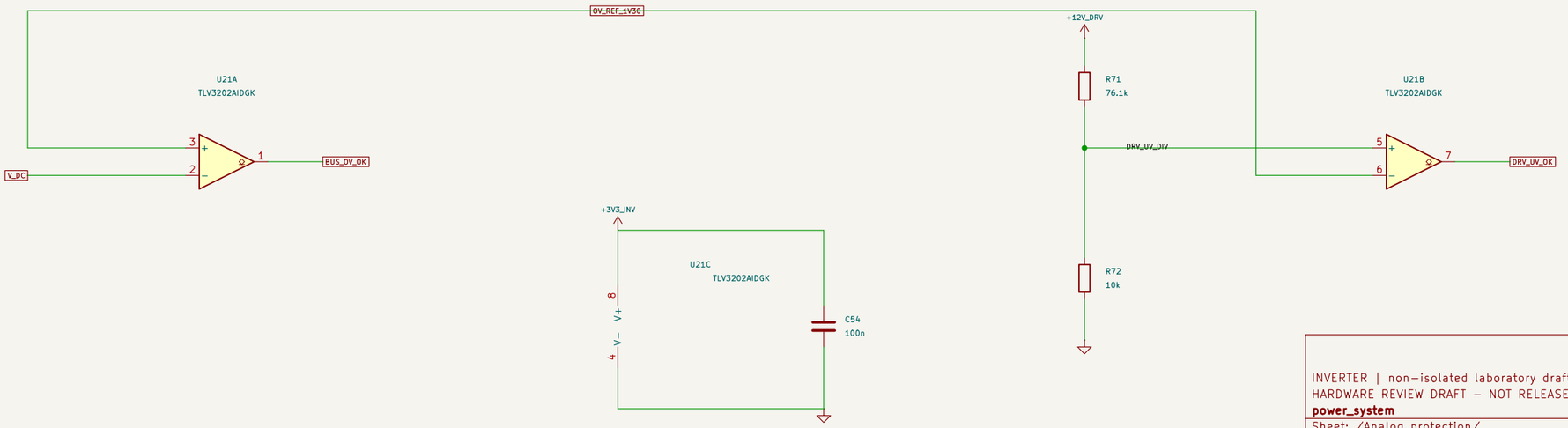
Size: A3	Date: 2026-11-06	**Rev: B-readable-draft**
KiCad E.D.A. 10.0.7		Id: 10/18

HARDWARE WINDOWS / EACH OUTPUT IS HEALTHY HIGH





BUS OVERVOLTAGE / DRIVER RAIL UNDERVOLTAGE



Healthy = HIGH. Window violation, bus >~65 V or driver rail <~11.193 V drives its own output LOW.
Push-pull outputs enter separate AND inputs. No wired-AND; no intentional hysteresis; first event is latched.

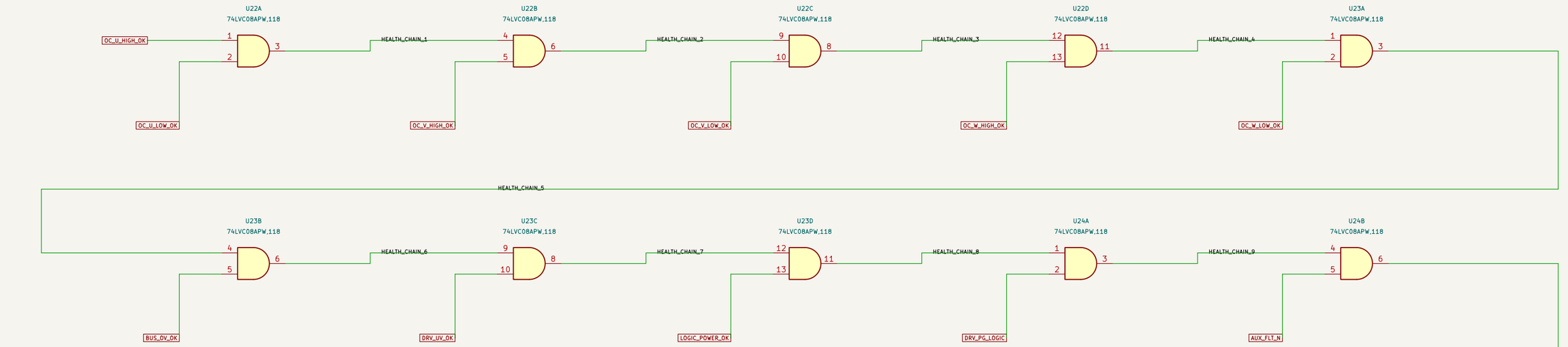
INVERTER | non-isolated laboratory draft | 12–60 V bus | separate 12–24 V auxiliary
HARDWARE REVIEW DRAFT — NOT RELEASED FOR PCB
power_system

Sheet: /Analog protection/
File: comparators.kicad_sch

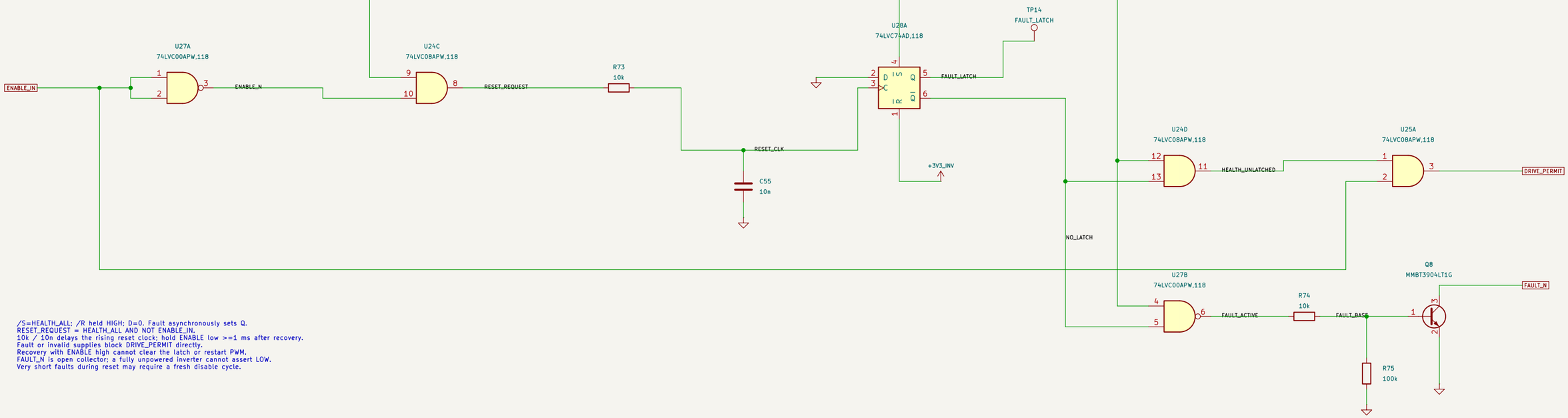
Title: Inverter — current, bus voltage and driver rail comparators

Size: A3	Date: 2026-11-06	**Rev: B-readable-draft**
KiCad E.D.A. 10.0.7		Id: 11/18

HARDWARE HEALTH CHAIN / EVERY INPUT MUST REMAIN HIGH



FAULT MEMORY / DELIBERATE DISABLE-TO-RESET / PERMISSION



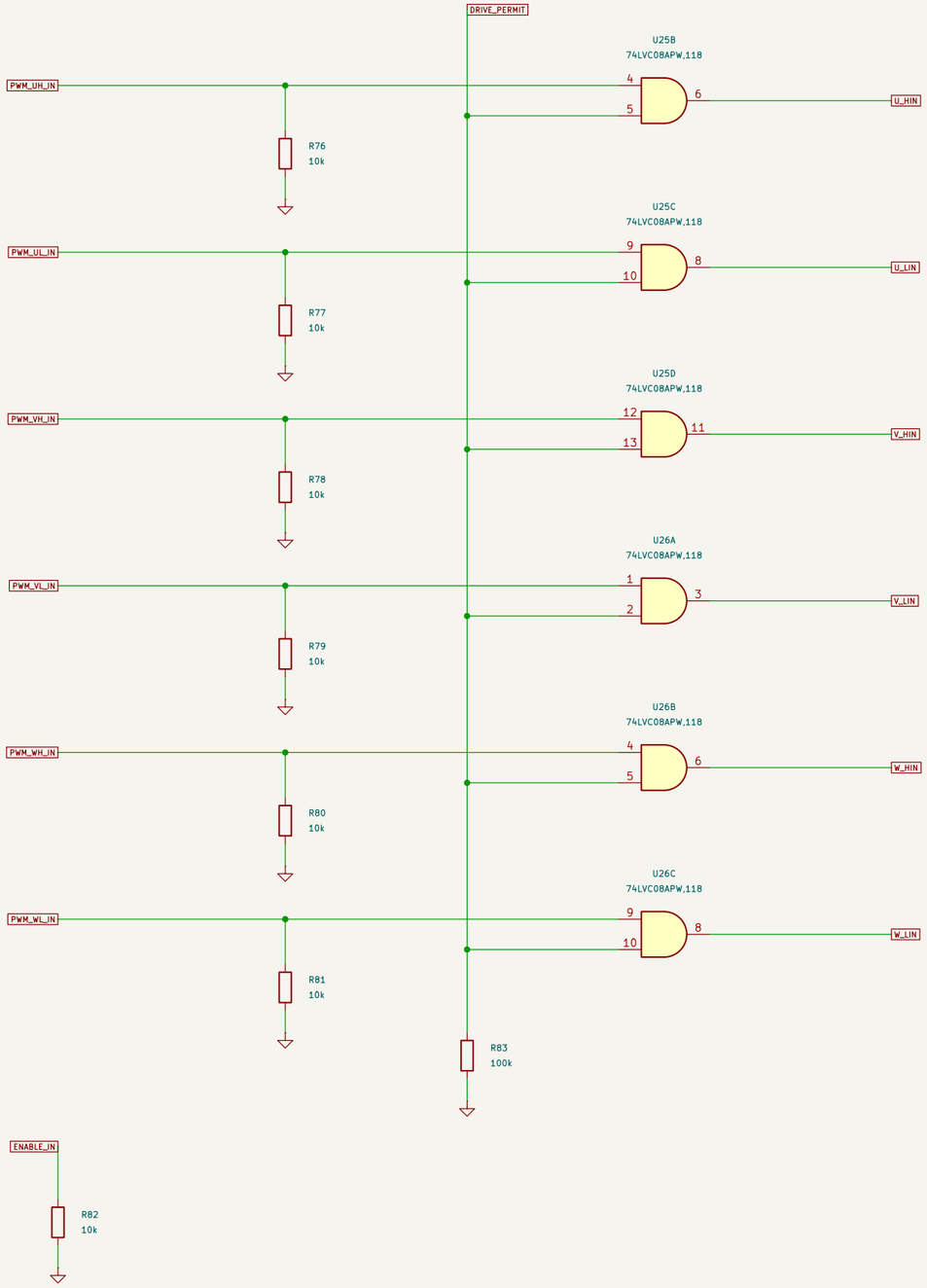
/S=HEALTH_ALL; /R held HIGH; D=0. Fault asynchronously sets Q.
RESET_REQUEST = HEALTH_ALL AND NOT ENABLE_IN.
10k / 10n delays the rising reset clock; hold ENABLE low >=1 ms after recovery.
Fault or invalid supplies block DRIVE_PERMIT directly.
Recovery with ENABLE high cannot clear the latch or restart PWM.
FAULT_N is open collector; a fully unpowered inverter cannot assert LOW.
Very short faults during reset may require a fresh disable cycle.

INVERTER | non-isolated laboratory draft | 12-60 V bus | separate 12-24 V auxiliary
HARDWARE REVIEW DRAFT - NOT RELEASED FOR PCB
power_system
Sheet: /Hardware latch/
File: fault_logic.kicad_sch

Title: Inverter - fault aggregation and reset latch

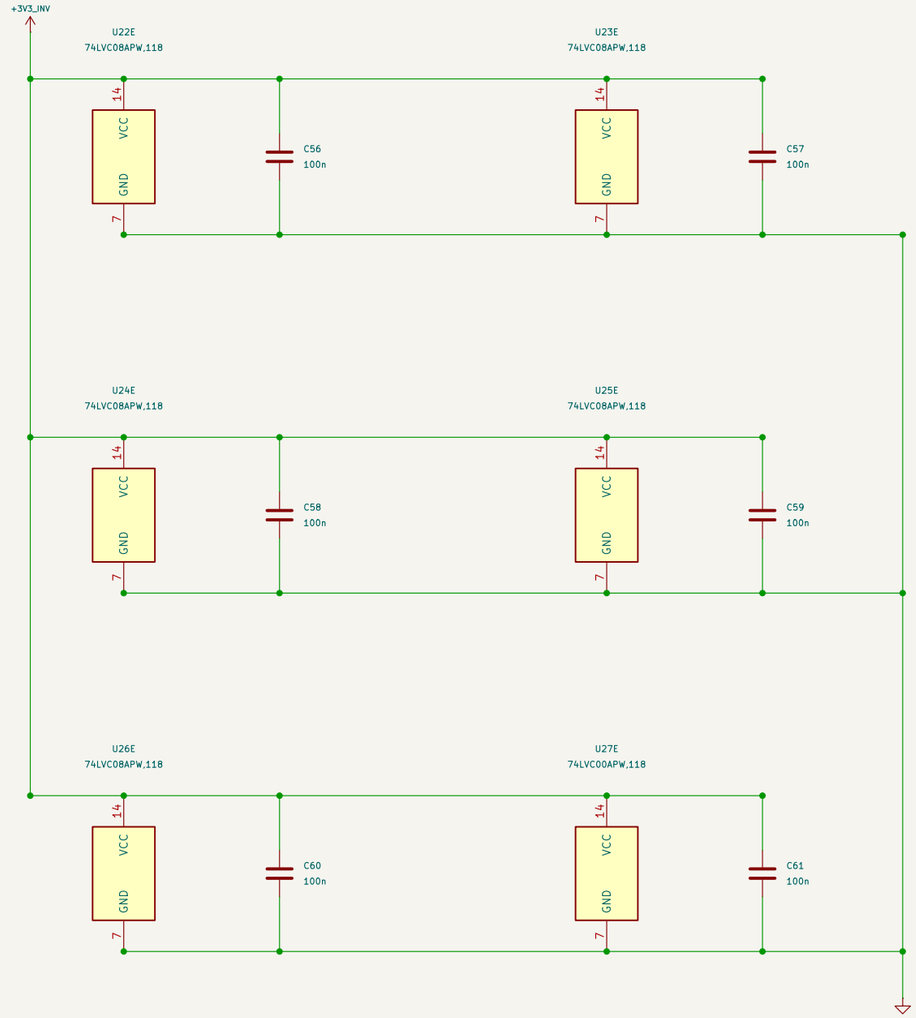
Size: A3	Date: 2026-11-06	**Rev: B-readable-draft**
KiCad E.D.A. 10.0.7		Id: 12/18

HARDWARE GATING / SIX PWM COMMANDS



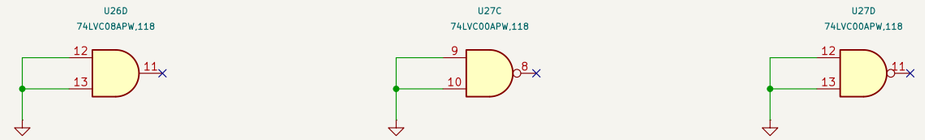
R76–R81: hold all six commands LOW while MCU outputs are Hi–Z.
R82: default disabled. R83: default no permission.
Logic levels remain 3.3 V; hardware AND gates operate independently of firmware.

LOGIC POWER / LOCAL DECOUPLING



C56–C61: 100 nF local bypass at each logic package.

UNUSED LOGIC / INPUTS TIED LOW



INVERTER | non–isolated laboratory draft | 12–60 V bus | separate 12–24 V auxiliary
HARDWARE REVIEW DRAFT – NOT RELEASED FOR PCB

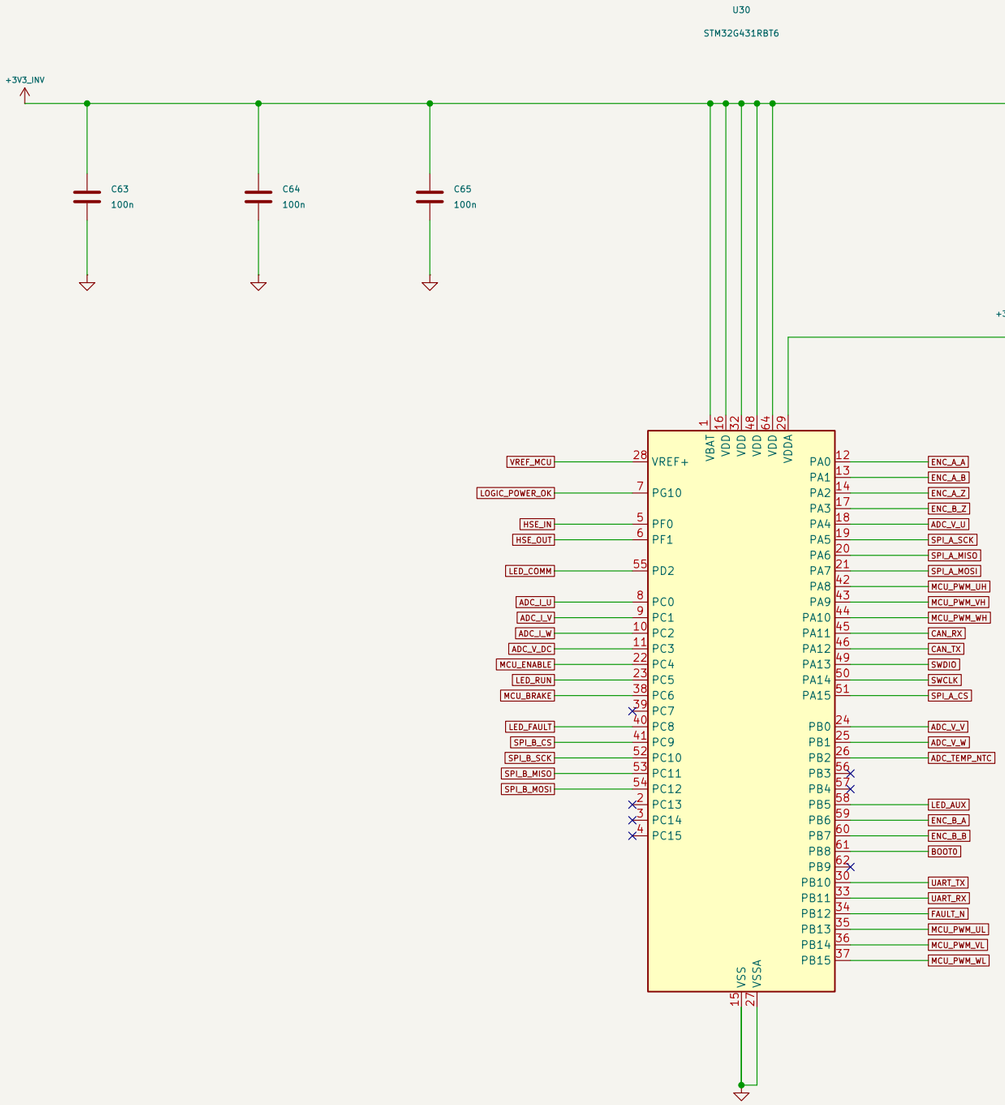
power_system

Sheet: /Hardware PWM gates/
File: control_interface.kicad_sch

Title: Inverter – local six PWM hardware gates

Size: A3	Date: 2026–11–06	**Rev: B–readable–draft**
KiCad E.D.A. 10.0.7		Id: 13/18

LOCAL STM32G431 / SUPPLY AND DECOUPLING



Power pins are wired to the local supply bus.
Each physical VDD and VBAT has its own 100 nF capacitor.
VSSA and VSS return to shunt LOW / DC−.
No external VCAP on this package.
VREF+ uses the external 3.3 V reference rail; VREFBUF OFF.

CROSS−SHEET SIGNALS
PWM / ENABLE / BRAKE −> local command and gate pages
ADC inputs <− local RC filters
ABZ / SPI <− external sensor interface
SWD / TTL / CAN / LEDs −> debug interface
Only these page boundaries use global signal labels.

Keep outputs disabled during reset and initialization. BKIN is active−low; disable automatic restart.
PB15 TIM1_CH3N uses AF4. Three currents share two ADCs: sampling windows still require validation.
Seven unused GPIOs remain explicit NC. Preserve NRST, SWD and the intended BOOT/UCPD configuration.

INVERTER | non−isolated laboratory draft | 12−60 V bus | separate 12−24 V auxiliary
HARDWARE REVIEW DRAFT − NOT RELEASED FOR PCB

power_system

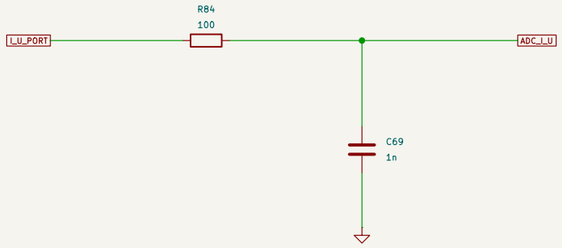
Sheet: /Local STM32G431/
File: mcu.kicad_sch

Title: Inverter − STM32G431 minimum system

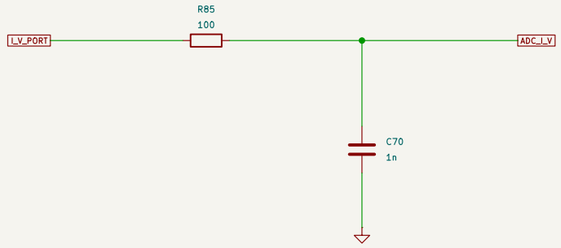
Size: A3	Date: 2026−11−06	**Rev: B−readable−draft**
KiCad E.D.A. 10.0.7		Id: 14/18

LOCAL ADC FILTERS / SOURCE -> SERIES R -> RC NODE -> MCU

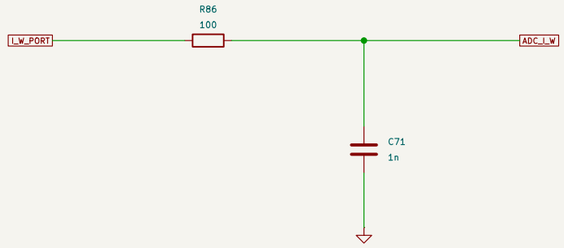
I_U



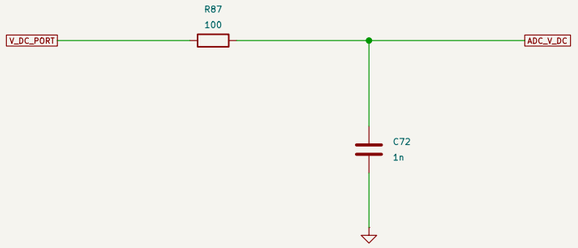
I_V



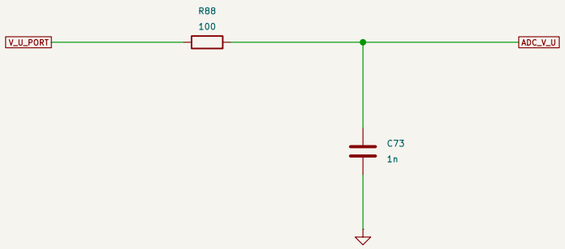
I_W



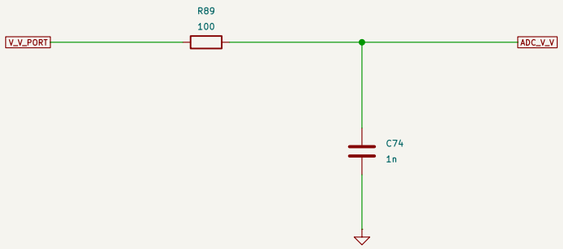
V_DC



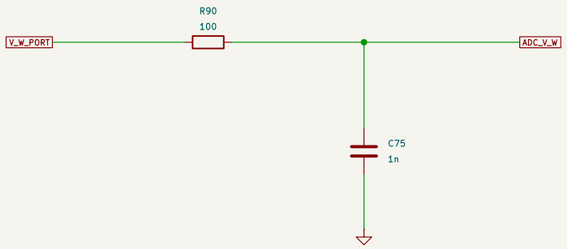
V_U



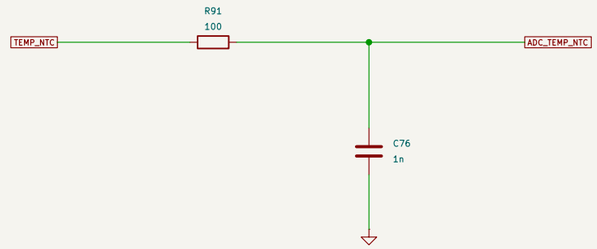
V_V



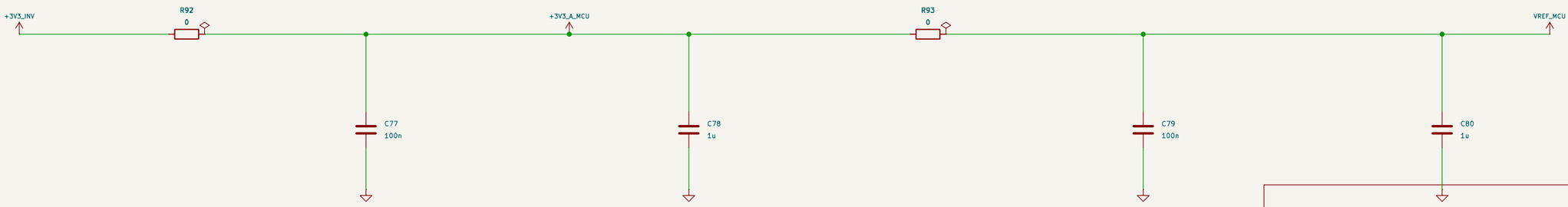
V_W



TEMP_NTC



VDDA AND VREF+ / EXTERNALLY SUPPLIED REFERENCE



Trip comparators remain upstream of the original 47R output isolators. New RC = 100R / 1nF.
NTC also has 10k / 10nF upstream; select adequate acquisition time. VREFBUF stays OFF.
The 1.65 V amplifier bias is not the ADC reference.

INVERTER | non-isolated laboratory draft | 12-60 V bus | separate 12-24 V auxiliary
HARDWARE REVIEW DRAFT - NOT RELEASED FOR PCB

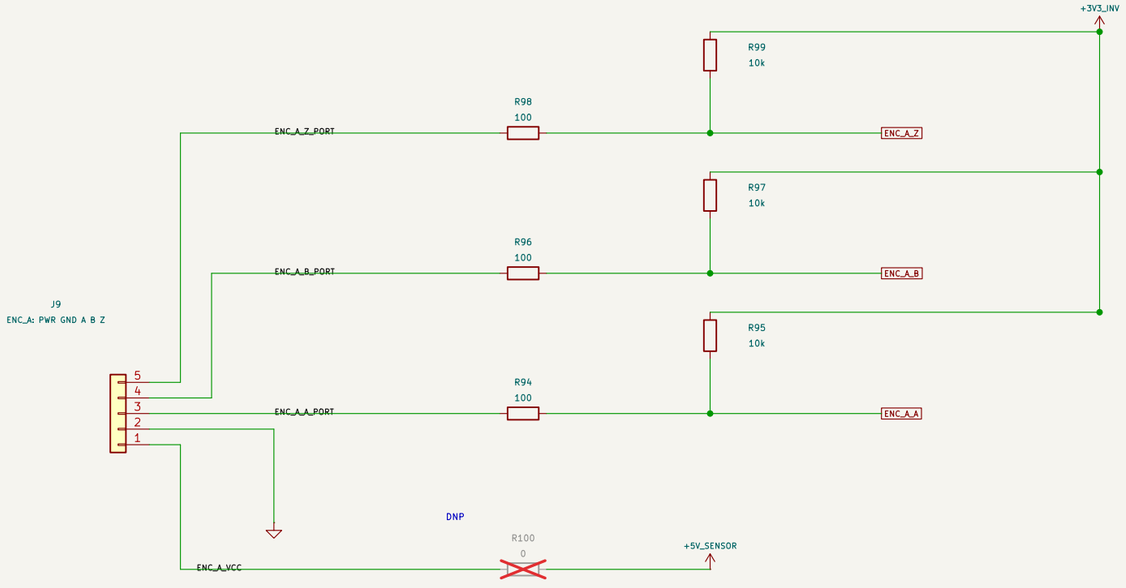
power_system

Sheet: /MCU analog inputs/
File: mcu_analog.kicad_sch

Title: Inverter - local ADC filters and analog supply

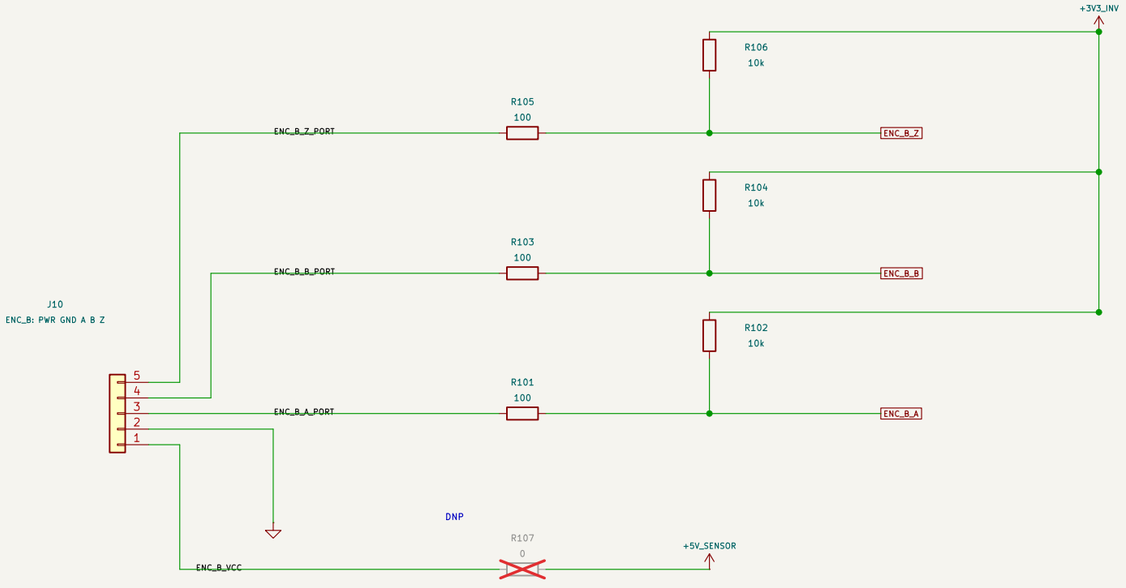
Size: A3	Date: 2026-11-06	**Rev: B-readable-draft**
KiCad E.D.A. 10.0.7		Id: 15/18

ABZ ENCODER A / 3.3 V SINGLE-ENDED



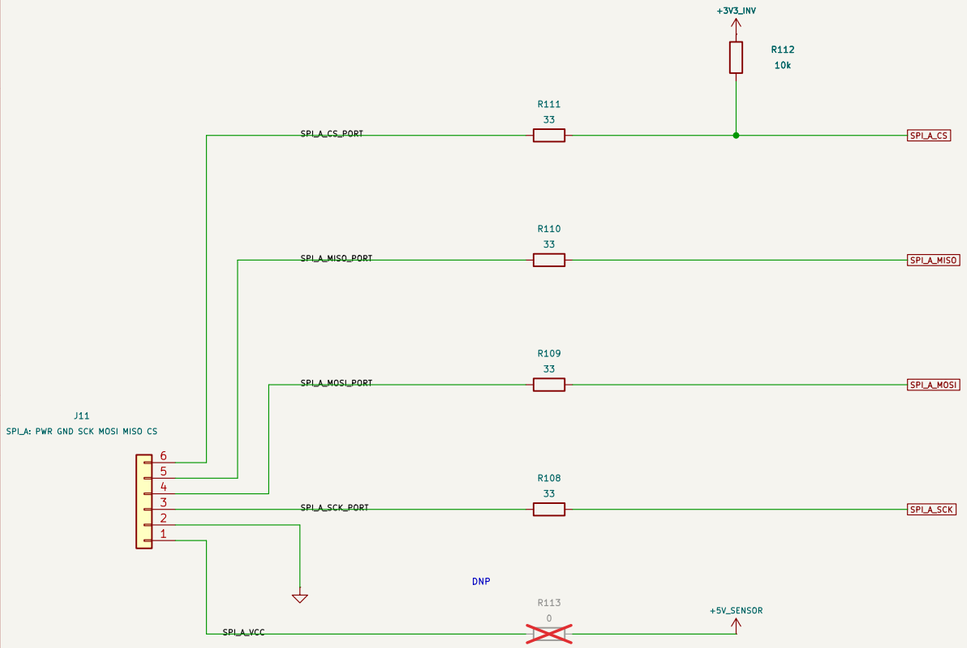
100 ohm series / 10k pull-ups to 3.3 V. 5 V supply link DNP.

ABZ ENCODER B / 3.3 V SINGLE-ENDED

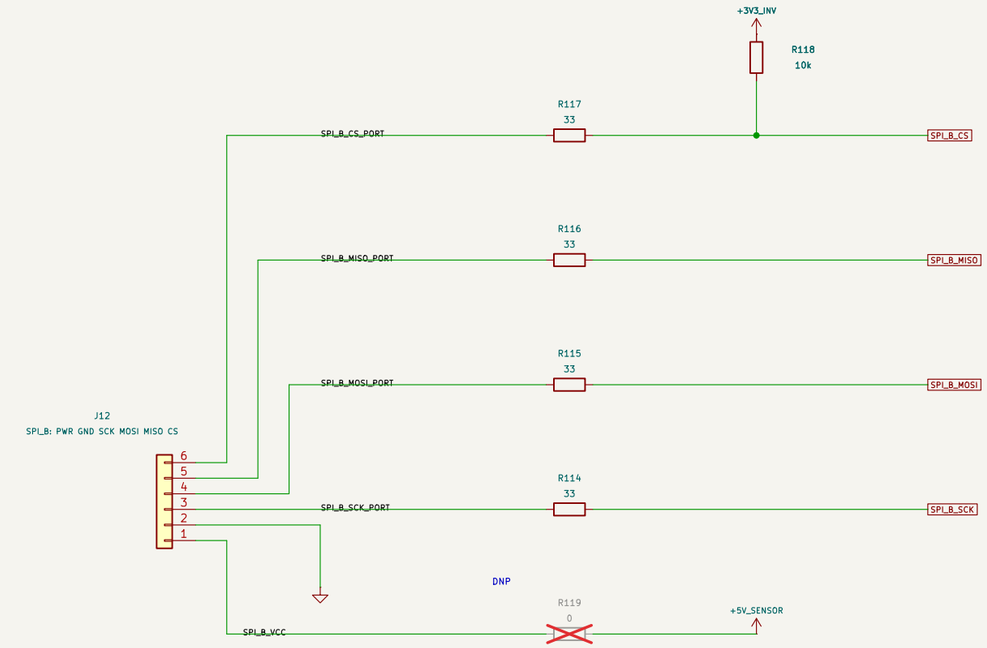


100 ohm series / 10k pull-ups to 3.3 V. 5 V supply link DNP.

SPI SENSOR A / 3.3 V SINGLE-ENDED



SPI SENSOR B / 3.3 V SINGLE-ENDED



All external signal pins are 3.3 V single-ended; no 5 V level conversion or cable protection is claimed.
R100 / R107 / R113 / R119 are DNP: external sensor-power links remain open by default.

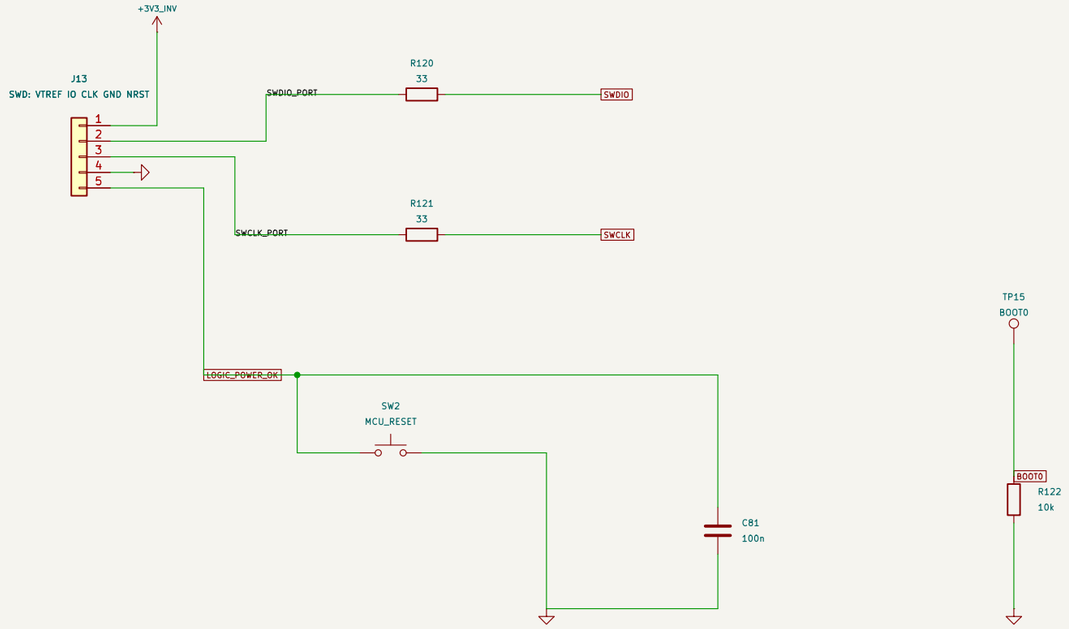
INVERTER | non-isolated laboratory draft | 12-60 V bus | separate 12-24 V auxiliary
HARDWARE REVIEW DRAFT - NOT RELEASED FOR PCB
power_system

Sheet: /Encoders and SPI/
File: sensors.kicad_sch

Title: Inverter - two ABZ and two SPI ports

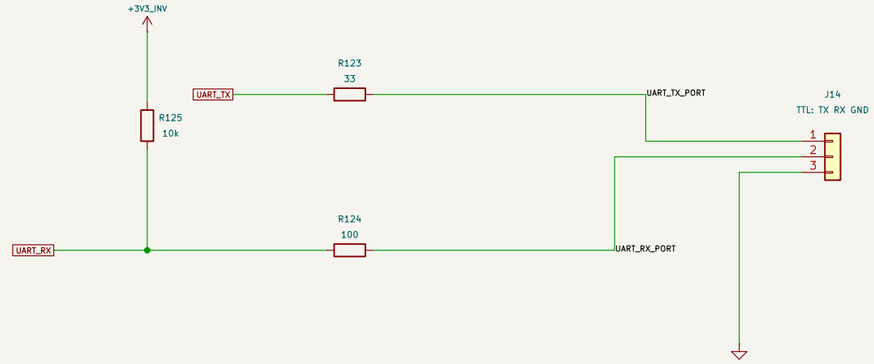
Size: A3	Date: 2026-11-06	**Rev: B-readable-draft**
KiCad E.D.A. 10.0.7		Id: 16/18

SWD / OPEN-DRAIN RESET / BOOT



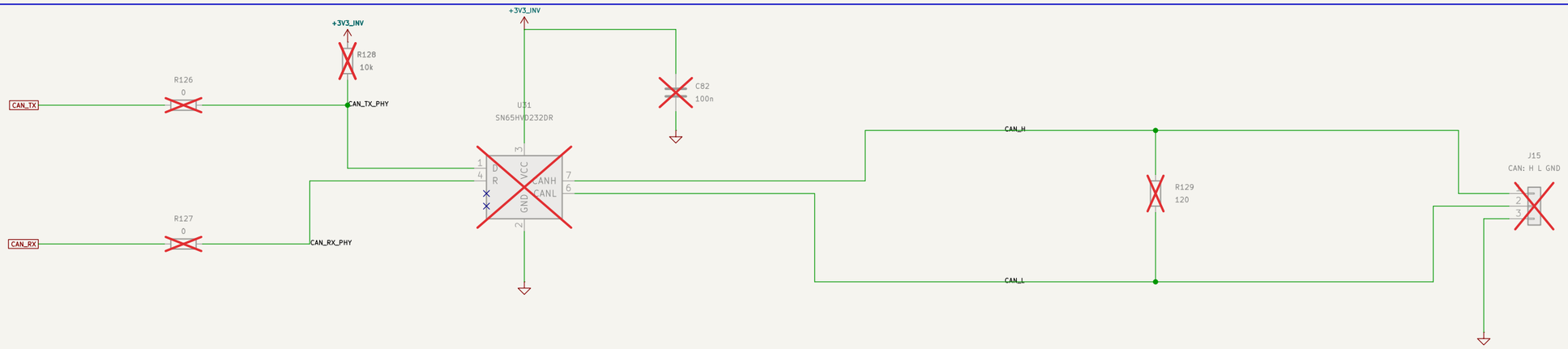
VTREF is sense only. NRST also enters the hardware health chain.
Ordinary power-stage faults do not reset the MCU.

TTL UART / USART3 / 3.3 V

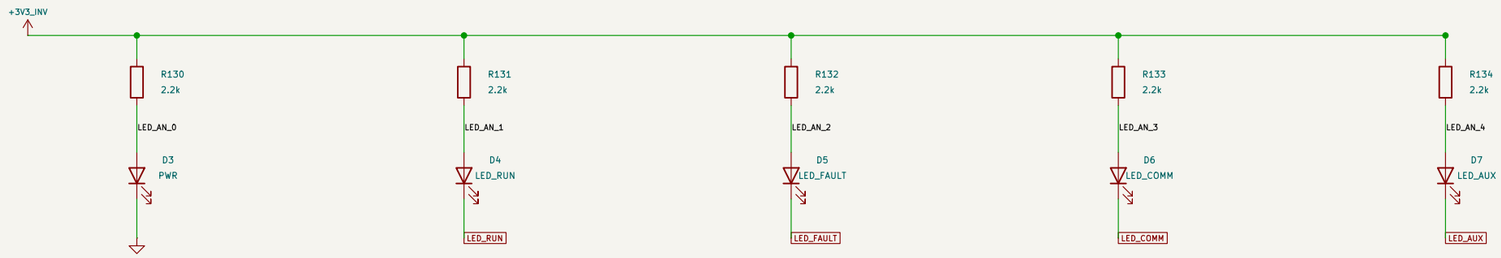


TX/RX relative to this board: no RS-232 levels.
No USB and no supply output on TTL connector.

OPTIONAL CLASSIC CAN / COMPLETE PHY BLOCK DNP



FIVE INDICATORS / ONE POWER + FOUR GPIO, ACTIVE LOW



INVERTER | non-isolated laboratory draft | 12-60 V bus | separate 12-24 V auxiliary
HARDWARE REVIEW DRAFT — NOT RELEASED FOR PCB
power_system

Sheet: /Debug and communication/
File: debug_comms.kicad_sch

Title: Inverter — SWD TTL optional CAN and status

Size: A3	Date: 2026-11-06	**Rev: B-readable-draft**
KiCad E.D.A. 10.0.7		Id: 17/18

LOCAL COMMAND SERIES RESISTORS / PAGE-TO-PAGE SIGNALS



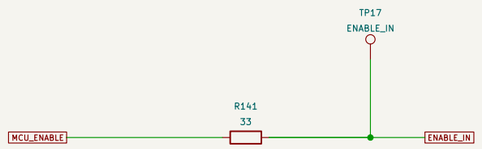






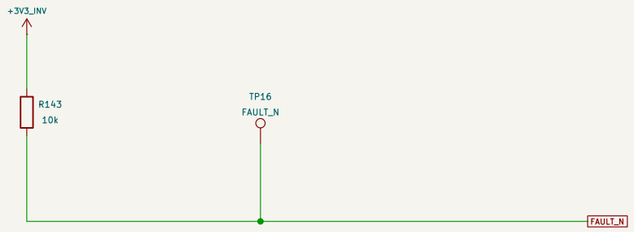






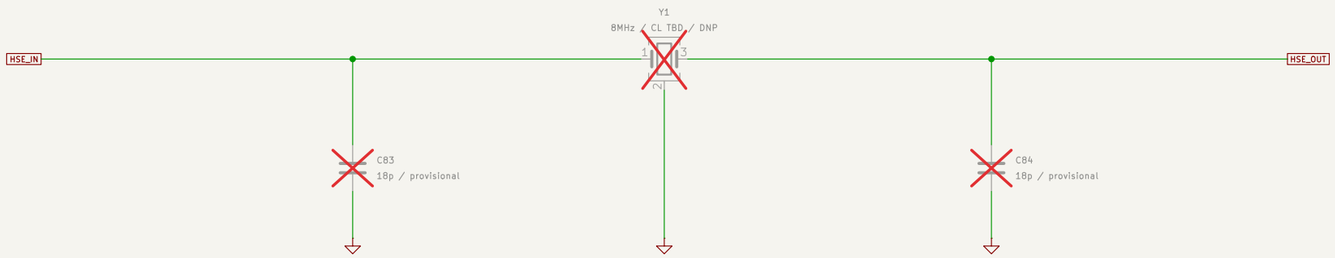


FAULT INPUT / LOCAL PULL-UP AND TIMER BKIN



Q8 open collector -> pull-up -> PB12 / TIM1_BKIN.
Bridge protection remains independent of firmware.
Hold ENABLE low >=1 ms before deliberate re-enable.
BRAKE remains independent of bridge ENABLE / fault.

HSE OPTION / CRYSTAL AND LOAD CAPACITORS DEFAULT DNP



Y1 8 MHz and load C are candidates.
Select actual CL / ESR / drive and
verify oscillator gain before fitting.
Initial bring-up can use HSI16 / PLL.

INVERTER | non-isolated laboratory draft | 12-60 V bus | separate 12-24 V auxiliary
HARDWARE REVIEW DRAFT - NOT RELEASED FOR PCB
power_system

Sheet: /Local command coupling/
File: mcu_control.kicad_sch

Title: Inverter - PWM enable fault brake and HSE

Size: A3	Date: 2026-11-06	**Rev: B-readable-draft**
KiCad E.D.A. 10.0.7		Id: 18/18